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Questions

Q1.

Many people swim as part of their regular exercise.

(a) The table gives some information about the effect of the intensity of exercise on heart rate.

Intensity of exercise / a.u.	Heart rate / bpm
4	118
6	120
8	138
10	150
12	155
14	160

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Describe the relationship between the intensity of exercise and heart rate.

(2)

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(b) Regular swimming affects resting heart rate and cardiac output.

cardiac output = stroke volume × heart rate

The table shows the mean resting cardiac output and mean resting stroke volume for a group of regular swimmers and a control group.

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Group	Mean resting cardiac output / $dm^3\ min^{-1}$	Mean resting stroke volume / cm^3
Regular swimmers	4.43	74.40
Control	4.21	58.40

Using the data in the table, calculate the difference in resting heart rate between these two groups.

Give your answer to 2 decimal places with appropriate units.

(3)

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Answer .....

*(c) Explain the role of the heart in responding to regular exercise.

Use the information in **both** tables and your own knowledge to support your answer.

(6)

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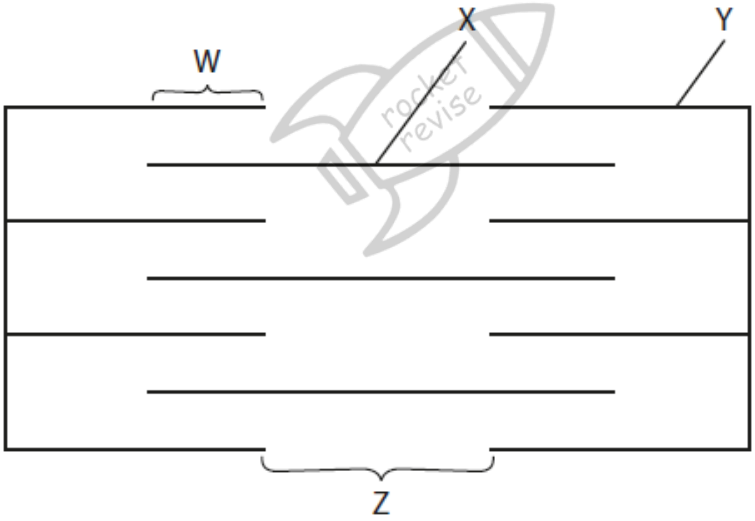
(Total for question = 11 marks)

Q2.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Skeletal muscles function to move the limbs of an organism.

(a) The diagram represents a sarcomere from skeletal muscle.



(i) Which protein forms the structure labelled X?

(1)

☒ **A** actin

☒ **B** myosin

☒ **C** tropomyosin

☒ **D** troponin

(ii) Which part becomes shorter when the muscle contracts?

(1)

☒ **A** W

☒ **B** X

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☒ **C** Y

☒ **D** Z

(iii) How many of the following statements are correct when a sarcomere contracts?

- calcium ions bind to troponin
- calcium ions are released from the sarcoplasmic reticulum
- myosin changes shape

(1)

☒ **A** none

☒ **B** one

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☒ **C** two

☒ **D** three

(iv) Which molecule contains an enzyme that hydrolyses ATP?

(1)

☒ **A** actin

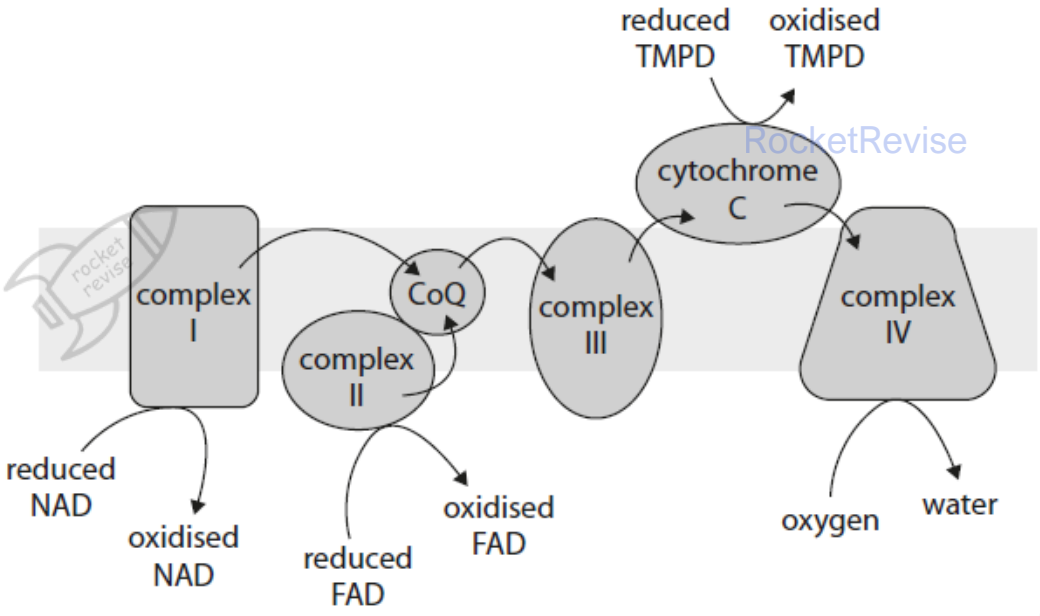
☒ **B** myosin

☒ **C** tropomyosin

☒ **D** troponin

*(b) Inherited mitochondrial diseases affect muscle function.

The diagram shows the electron transport chain in a mitochondrion.



In an investigation, samples of muscle tissue from healthy individuals and from four individuals with mitochondrial disease were collected.

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The function of mitochondria, when these samples were treated with different substrates, was investigated.

Treating muscle tissue from healthy individuals with:

- pyruvate and malate increases the production of reduced NAD
- succinate increases the production of reduced FAD
- reduced TMPD increases the reduction of cytochrome c.

The oxygen consumption of the muscle tissue samples from healthy individuals and from four individuals with mitochondrial disease were recorded.
The results are shown in the table.

Sample taken from	Oxygen consumption in the presence of each treatment / pmol O_2 min^{-1} mg^{-1} of muscle tissue		
	pyruvate and malate	succinate	reduced TMPD
Individual 1	3.0	3.5	13.0
Individual 2	18.0	10.0	60.0
Individual 3	6.5	14.0	51.0
Individual 4	7.5	6.0	40.0
Healthy individuals	21.0 ± 5.0	13.5 ± 4.5	63.0 ± 20.0

The results of this investigation can be used to identify the defects in the electron transport chain of individuals with mitochondrial disease.

Comment on the defects in the electron transport chain in the individuals with mitochondrial disease.

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Q3.

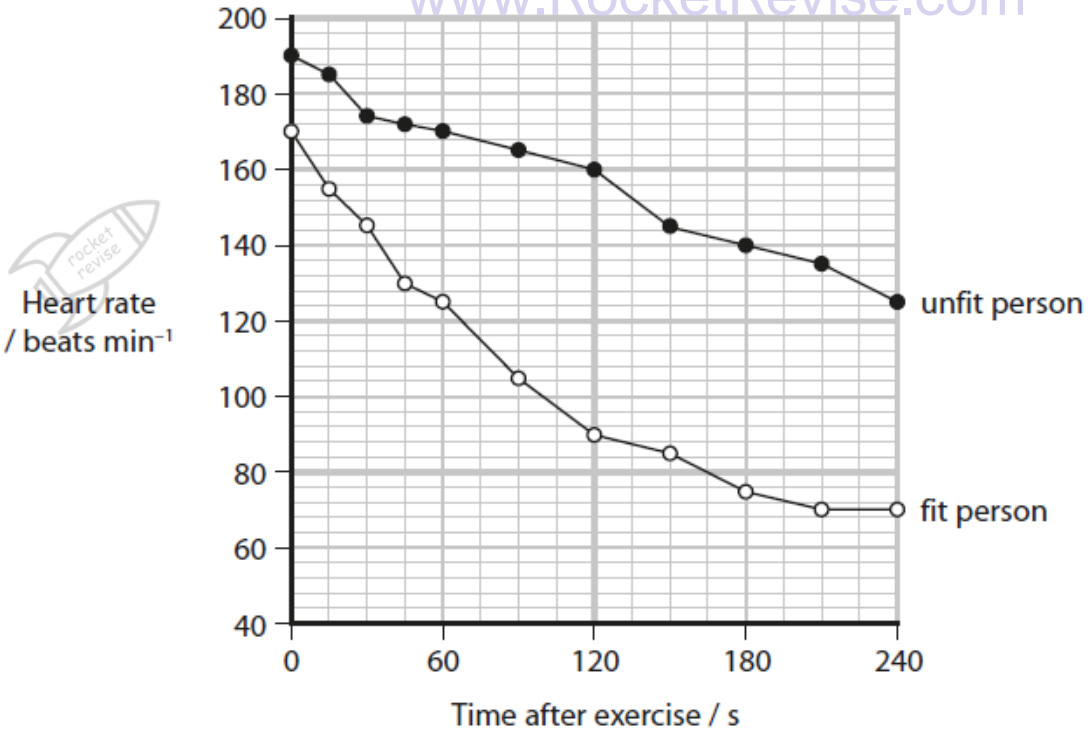
Exercise affects the cardiac output of a person.

The graph shows the change in heart rate following exercise for an unfit person and for a fit person.

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(a) The heart rate of the unfit person decreased by 16%, 120 seconds after exercise.

Calculate the difference in percentage decrease in heart rate of these two people, 120 seconds after exercise.

Give your answer to an appropriate number of decimal places.

(3)

Answer .....

(b) Calculate the heart rate of the unfit person at 5 minutes.

The change in heart rate of the unfit person is a linear relationship.
Use the equation for a straight line y = mx + c.

(3)

Answer .....

(c) Explain the difference in the time it would take for these two individuals to recover from the exercise.

(3)

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(d) Describe how heart rate is controlled in response to changes in the blood, following exercise.

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(Total for question = 13 marks)

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Q4.

Researchers investigated how groups of rats, treated with drugs, responded to stress.

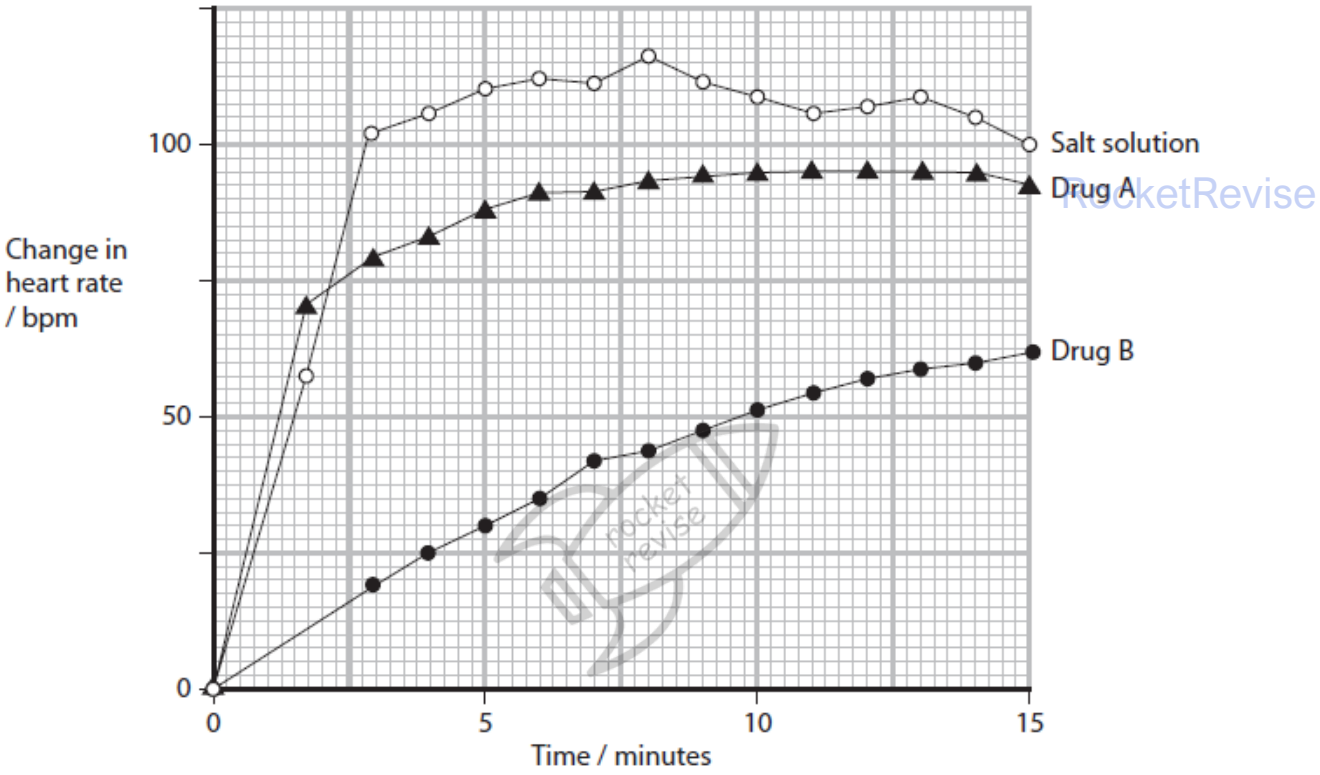
Groups of rats were stressed by moving them into a new environment. The rats were then treated with a salt solution, drug A or drug B.

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Group	Treatment
1	Salt solution
2	Drug A
3	Drug B

The graph shows the change in mean heart rate measured over a 15-minute period.



(a) Calculate the percentage difference in the change in heart rate when using drug B between 5 and 12.5 minutes.

(3)

Answer ..... %

(b) Compare and contrast the effects of drug A and drug B on the heart rate during the investigation.

(3)

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(c) Explain the effect of these drugs on the control of the heart rate.

(3)

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(Total for question = 9 marks)

Q5.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

The kidney is involved in the process of homeostasis in a mammal.

(a) Antidiuretic hormone (ADH) is required for homeostasis.

How many of these parts of the kidney respond to ADH?

- Bowman's capsule
- collecting duct
- loop of Henle
- proximal tubule

(1)

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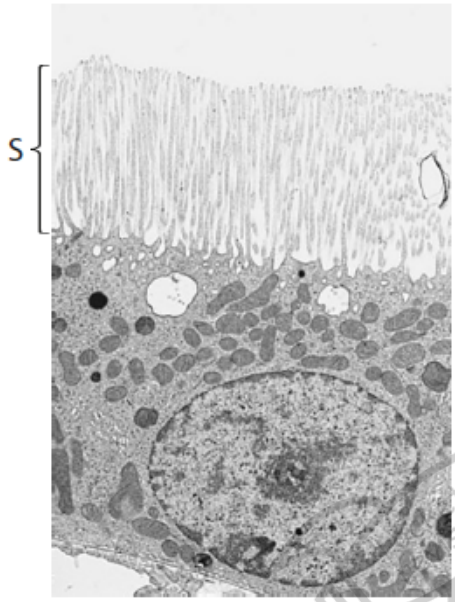
☒ **A** 1

☒ **B** 2

☒ **C** 3

☒ **D** 4

(b) The electron micrograph shows a transverse section of part of a proximal tubule.



(Source: https://veteriankey.com/solute-reabsorption/)

(i) Name the process by which the fluid moves from the plasma into the renal tubule (nephron).

(1)

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(ii) Name the structures labelled S.

(1)

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(c) A student compared the left renal veins of two individuals.

(i) The diameter of the left renal vein of one individual was 4.71 mm.

Calculate the cross-sectional area of this vein.

(2)

Answer ..... mm^2

(ii) The cross-sectional area of the left renal vein of the second individual was 0.194 cm^2.

Calculate the difference in the cross-sectional area in the left renal vein of these two individuals.

(1)

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Answer .....

(d) The table shows the concentration of glucose and urea in the blood plasma, glomerular filtrate and urine of an individual.

Substance	Concentration in blood plasma / g dm^{-3}	Concentration in glomerular filtrate / g dm^{-3}	Concentration in urine leaving collecting duct / g dm^{-3}
glucose	1.50	1.50	0.00
urea	0.30	0.30	21.00

Explain the concentration of glucose and urea in the urine.

Use the information in the table to support your answer.

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(e) An individual produced 0.83 cm^3 of urine each minute.

Calculate the volume of urine produced by this individual in 24 hours.

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Assume the urine production is at a constant rate.

Give your answer in dm^3, to two significant figures.

(2)

Answer ..... dm^3

(Total for question = 11 marks)

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Q6.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The human body responds to changes in both its internal and external environments.

(a) How many of these statements describe the effects of adrenaline in the "fight or flight" response?

- arterioles constrict in skeletal muscles
- glycogen is released from the liver
- the rate of food digestion decreases

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(1)

☒ **A** 0

☒ **B** 1

☒ **C** 2

☒ **D** 3

(b) The heart responds to adrenaline by increasing cardiac output.

Cardiac output = heart rate × stroke volume

(i) Calculate the cardiac output for a person who has a heart rate of 77 bpm and a stroke volume of 70 cm^3.

Give your answer to two significant figures in dm^3 $hour^{-1}$.

(2)

Answer ..... dm^3 hour–1

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(ii) The formula shows how to calculate the cardiac index. This is a method of adjusting cardiac output values for the body surface area of an individual.

$$\text{cardiac index} = \frac{\text{cardiac output in dm}^3\,\text{min}^{-1}}{\text{body surface area in m}^2}$$

The table shows some information for cardiac output, body surface area and cardiac index for two individuals.

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Individual	Cardiac output / $dm^3 min^{-1}$	Body surface area / m^2	Cardiac index
A	6.7	1.92	
B		1.49	2.8

Complete the table to show the units for cardiac index, the cardiac index for individual A and the cardiac output for individual B.

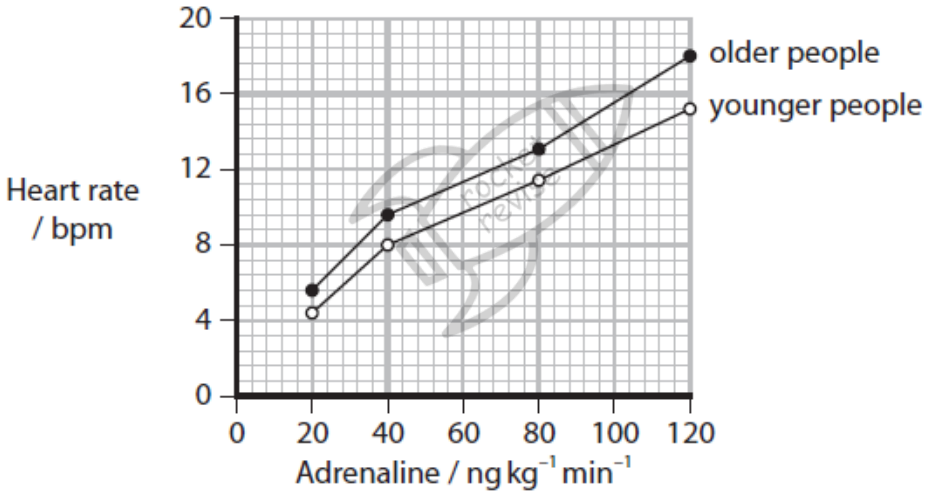
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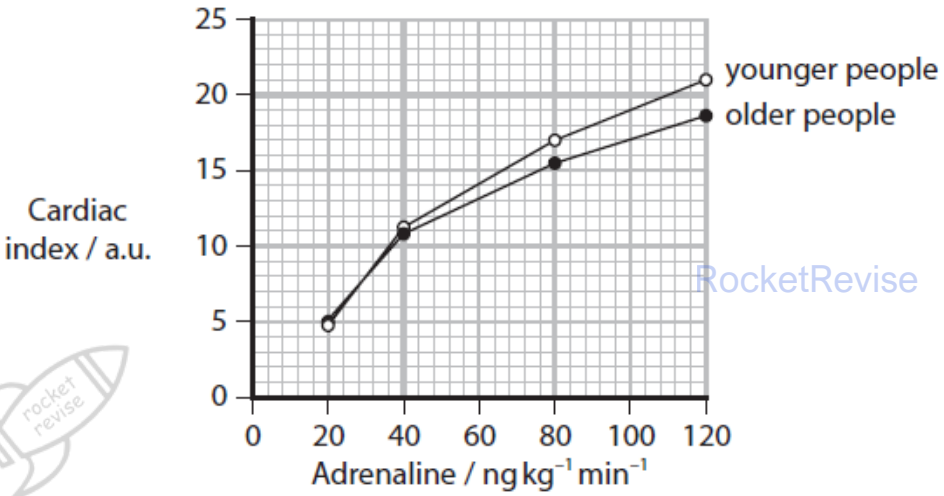
*(iii) Ageing is associated with changes in the way the heart responds to adrenaline.

In an investigation, the effects of adrenaline on the hearts of young people and old people were recorded.

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The graphs show the results of this investigation.

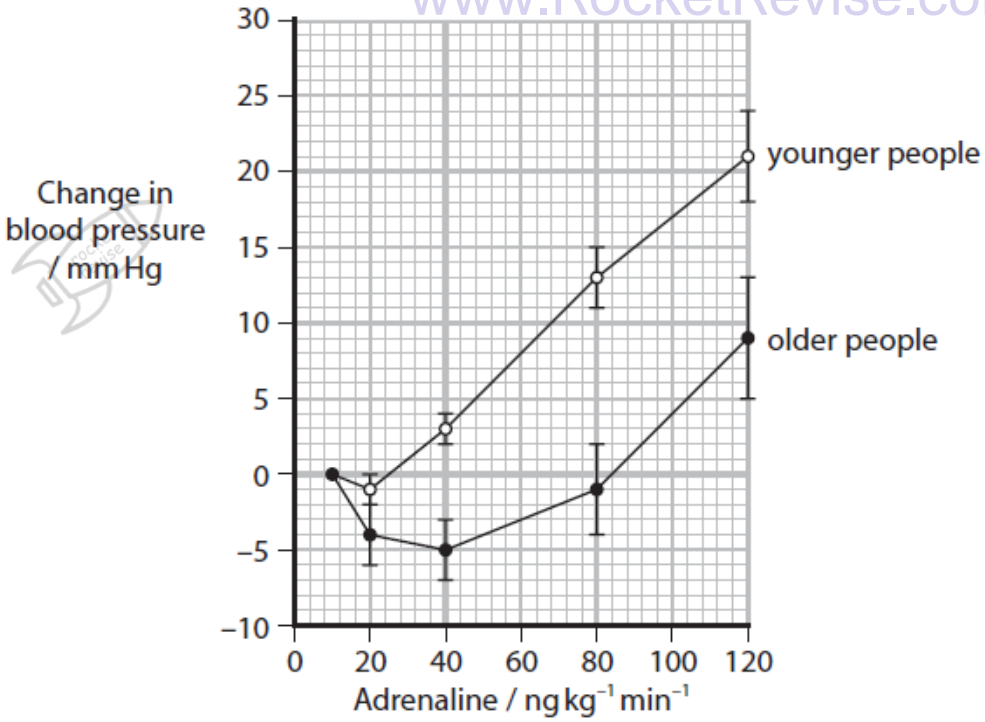




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Discuss the effect of ageing on the response of the heart to adrenaline.

Use the information in the graphs to support your answer.

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(Total for question = 12 marks)

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Q7.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

In eukaryotic organisms aerobic respiration takes place in mitochondria.

(a) The diagram shows a mitochondrion.



The actual length of the mitochondrion between A and B was 0.75 µm.

Calculate the magnification of the mitochondrion shown in the diagram.

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Give your answer in standard form.

(2)

Answer .....

(b) In an investigation, the rate of production of carbon dioxide by mitochondria was measured.

Three samples of mitochondria had the following chemicals added to them:

- Sample 1: pyruvate
- Sample 2: pyruvate and ADP
- Sample 3: pyruvate and pyruvate transport inhibitor.

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The table shows the results of this investigation.

Sample	Chemicals added	Rate of carbon dioxide production in 20 minutes / nmol per mg of mitochondria
1	pyruvate	62.6
2	pyruvate and ADP	85.0
3	pyruvate and pyruvate transport inhibitor	11.6

(i) Which is the approximate rate at which carbon dioxide is produced in sample 1?

(1)

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☒ **A** 3 nmol mg min

☒ **B** 180 nmol mg min

☒ **C** 3 nmol mg^{-1} min^{-1}

☒ **D** 180 nmol mg^{-1} min^{-1}

(ii) Explain the results of this investigation.

Use the information in the table and your own knowledge of the Krebs cycle to support your answer.

(4)

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(c) Fast twitch and slow twitch muscle fibres contain different numbers of mitochondria.

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(i) Explain the difference in the number of mitochondria in these two types of muscle fibre.

(3)

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(ii) Describe the role of ATP in muscle contraction.

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(Total for question = 14 marks)

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Q8.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Muscles, bones and joints enable movement of the skeleton.

(a) (i) Which structure attaches one bone to another bone at a flexible joint?

(1)

☒ **A** actin

☒ **B** ligament

☒ **C** synapse

☒ **D** tendon

(ii) How many of the statements about joints are correct?

(1)

- antagonistic pairs of muscles move the bones at a joint
- muscles acting on a joint are myogenic
- tendons are more elastic than ligaments

☒ **A** 0

☒ **B** 1

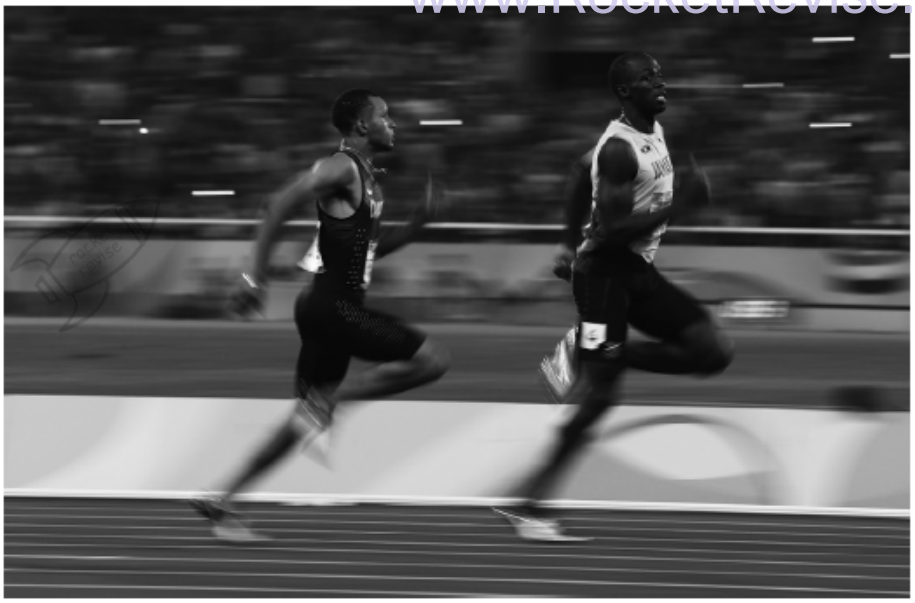
☒ **C** 2

☒ **D** 3

(b) Usain Bolt's world record of 9.58 seconds for the 100 m men's sprint is unbeaten.

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(Source: © Ian MacNicol / Getty Images)

(i) What is the role of myoglobin in muscle fibres?

(1)

☒ **A** it is an enzyme that reacts with myosin

☒ **B** it is part of the sarcolemma involved in the release of calcium ions

☒ **C** to act as an immediate source of energy

☒ **D** to provide oxygen for muscles

(ii) Explain why the muscles of a sprinter have a high percentage of fast twitch fibres.

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(c) In an investigation, an athlete ran for 10 minutes on a running machine moving at constant speed.

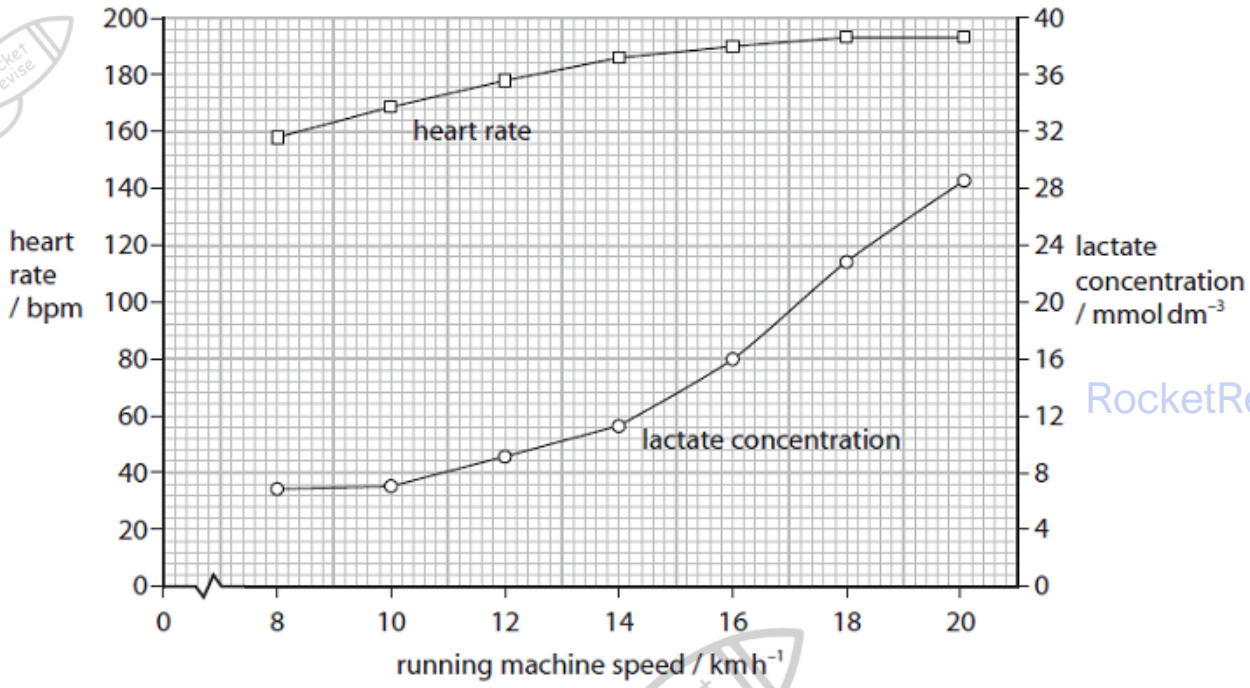
The investigation was repeated at seven running speeds.

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At each speed, the heart rate and lactate concentration in the blood of the athlete were measured.

The results are shown in the graph.



Comment on the results of this investigation.

(3)

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(Total for question = 9 marks)

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Q9.

Animal species have different resting heart rates.

(a) The table shows the resting heart rate and mean life expectancy for some animal species.

Animal species	Resting heart rate / bpm	Mean life expectancy / years	Mean mass / kg	Mean resting metabolic rate / $dm^3\ O_2\ kg^{-1}\ hr^{-1}$
elephant	30	80	5780	1400
horse	38	30	500	910
human	72	78	110	360
rabbit	205	12	7	48
rat	420	1.8	2	10.8
humming bird	1260	7	0.02	0.012

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(i) Describe the relationships between resting heart rate and the data shown in the table.

(2)

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(ii) Calculate the volume of oxygen produced each hour by a resting elephant with a mass of 4500 kg.

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Give your answer in standard form.

(3)

Answer .....

(b) The photograph shows a cheetah.

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(Source: © Stu Porter/Shutterstock)

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The heart rate for a cheetah increases from 120 bpm to 250 bpm during a chase. It can reach speeds of 100 km hr^{-1}

(i) Explain how the heart rate of a cheetah can be increased during a chase.

(4)

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(ii) Suggest why a cheetah can maintain this heart rate and speed for only a short period of time.

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(Total for question = 11 marks)

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Q10.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The giraffe is the tallest mammal and has a long neck.

The photograph shows a giraffe.



(Source: © Eugen Haag/Shutterstock)

(a) (i) Which structure joins the bones of the neck together?

(1)

☒ **A** actin

☒ **B** ligament

☒ **C** muscle

☒ **D** tendon

(ii) Which structure joins a muscle to a bone?

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(1)

☒ **A** actin

☒ **B** ligament

☒ **C** myosin

☒ **D** tendon

(iii) Which chemical group contains actin and myosin?

(1)

☒ **A** carbohydrates

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☒ **B** fatty acids

☒ **C** nucleic acids

☒ **D** proteins

(b) Giraffes must bend down to drink water.



(Source: © Wim Hoek/Shutterstock)

(i) Explain how the knee joint can be held steady in this position as the giraffe drinks.

(2)

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(ii) Describe the role of ATP in the sliding filament theory of muscle contraction.

(3)

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(Total for question = 8 marks)

Q11.

The ventilation rate of animals will depend on many factors.

(a) State **two** lung conditions that affect the ventilation rate of a human.

(2)

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(b) The table shows the mean resting ventilation rate of five species of animal.

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Animal	Body mass / kg	Resting ventilation rate / breaths per second
Talapoin monkey	1.3	0.54 ± 0.09
Ring-tailed lemur	2.9	0.53 ± 0.04
Crested porcupine	20.0	0.30 ± 0.03
Siberian tiger	134.0	0.27 ± 0.06
Hippopotamus	2210.0	0.12 ± 0.03

(i) Comment on the relationship between the mean resting ventilation rate and the body mass of animals.

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(3)

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(ii) The tidal volume for a resting talapoin monkey was found to be 40 cm^3.

The minute ventilation is the volume of gas inhaled or exhaled from the lungs in one minute.

Calculate the respiratory minute ventilation for a resting talapoin monkey.

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Give your answer in dm^3 to **two** significant figures.

(2)

..... dm^3

(c) An increase in activity level results in a change in the ventilation rate.

Describe why this change occurs and how it is controlled.

(4)

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(Total for question = 11 marks)

Q12.

The concentration of ions in the blood is controlled by homeostasis.

(a) Explain how ions can cross the cell membrane.

(3)

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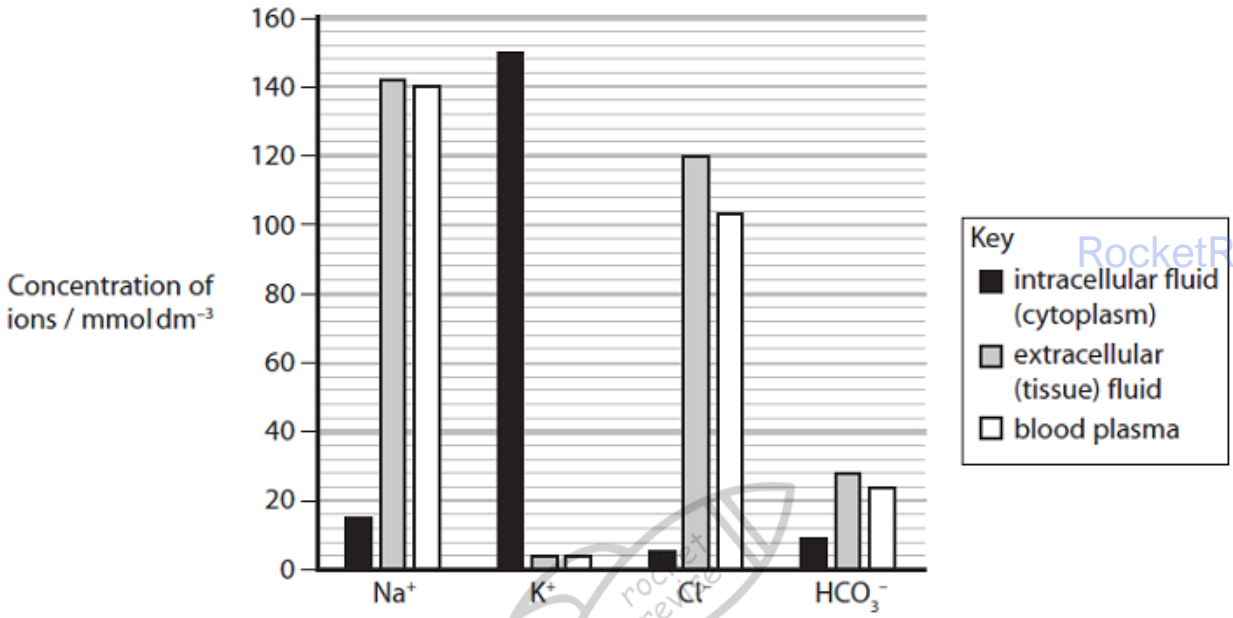
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*(b) Ions are found within fluids in the human body in different concentrations.

The graph shows the concentration of ions in intracellular fluid (cytoplasm), extracellular (tissue) fluid) and blood plasma.



The ions are used to perform vital functions within the metabolism of the cells.

Discuss how the distribution and concentrations of the ions shown in the graph contribute to their biological roles.

Use the information given in the graph and your own knowledge to support your answer.

(6)

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(Total for question = 9 marks)

Q13.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Skeletal muscle contains slow twitch and fast twitch muscle fibres.

(a) The table compares the percentage of slow twitch fibres in different types of athlete.

Type of athlete	Range of percentages of slow twitch fibres in muscle (%)
100 m runner	12 to 30
1500 m runner	30 to 65
10 000 m runner	60 to 95
marathon (42 200 m) runner	60 to 95

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Explain the importance of slow twitch muscle fibres in these types of athlete.

(3)

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(b) Name the basic unit of contraction in a skeletal muscle.

(1)

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(c) The heart pumps oxygen rich blood to the tissues.

Cardiac output and cardiac index can show how effectively the heart is working.

(i) A patient had a heart rate of 65 beats min^{-1} and a stroke volume of 57 cm^3 per beat.

Calculate the cardiac output for this individual.

Use the equation:

stroke volume = cardiac output ÷ heart rate

Give your answer in dm^3 min^{-1}.

(2)

Answer .....

The cardiac index can be calculated using the formula:

$$\text{cardiac index} = \frac{\text{stroke volume} \times \text{heart rate}}{\text{body surface area}}$$

The table shows data for two patients.

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Patient	heart rate /bpm	stroke volume /dm^3	body surface area /m^2
A	79	0.057	1.54
B	79	0.075	2.39

(ii) The cardiac index for patient A was 2.92 dm^3 min^{-1} m^{-2}

Which is the correct cardiac index for patient B?

(1)

☒ **A** 1.24

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☒ **B** 2.48

☒ **C** 2.92

☒ **D** 5.21

(iii) State the ratio of the body surface area of patient A to patient B.

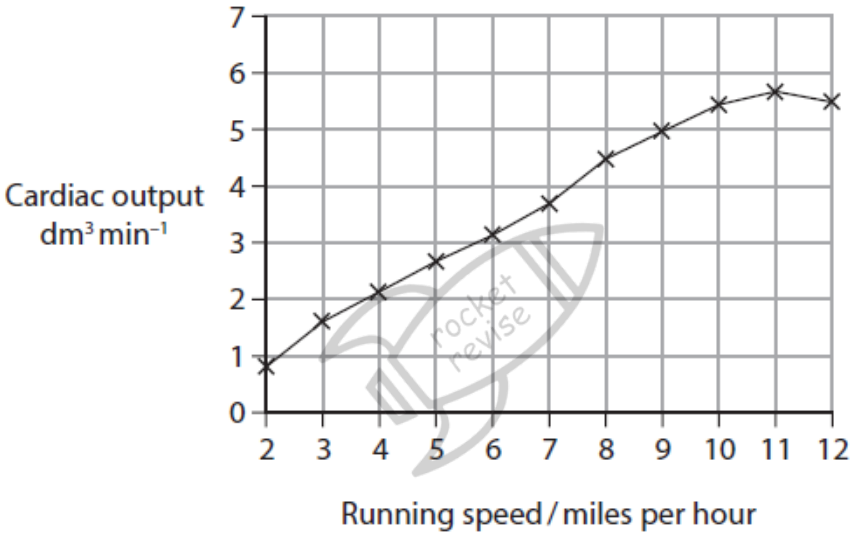
(1)

Answer .....

(d) The cardiac output changes in an athlete as their running speed increases.

The graph shows cardiac output for an athlete running at different speeds.

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(i) Comment on the effect of running speed on cardiac output.

(2)

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(ii) Exercise increases the rate of respiration in muscles.

Explain how an increase in the rate of respiration causes an increase in the heart rate of an

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athlete.

(3)

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(iii) Anaerobic respiration produces lactate.

Describe what happens to this lactate.

(2)

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(Total for question = 15 marks)

Q14.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The mean resting heart rate varies between species of mammal.

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The table shows the mean resting heart rate and mean lifespan for seven mammals.

Mammal	Mean resting heart rate / bpm	Mean lifespan / years
mouse	595	1
monkey	275	15
cat	120	17
giraffe	82	19
human	72	80
elephant	41	23
whale	15	29

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(a) Describe **two** conclusions that can be drawn from these data.

(2)

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(b) An electrocardiogram (ECG) is used to identify abnormal heart rhythms.

Trace A shows a normal heart rhythm and trace B shows an abnormal heart rhythm.



(i) Which is the heart rate shown in trace A?

(1)

☒ **A** 6 bpm

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☒ **B** 48 bpm

☒ **C** 60 bpm

☒ **D** 75 bpm

(ii) Describe **two** differences shown between trace A and trace B.

(2)

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(c) Cardiac muscle is unable to regenerate or repair itself. Mammalian heart muscle can beat only a certain number of times in a lifetime.

Humans have a mean resting heart rate of 72 beats per minute and a mean lifespan of 80 years.

(i) Suggest **two** reasons why humans can have a different lifespan from other mammals.

(2)

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(ii) State why cardiac output provides more information than the resting heart rate.

(1)

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(iii) An African giraffe has the longest stride of all land mammals and can run at a speed of up to 60 km hr^{-1}.

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Describe how the nervous system affects cardiac output to enable the giraffe to run at this maximum speed.

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(Total for question = 12 marks)

Q15.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Movement in mammals results from interactions between muscles, tendons, ligaments and bones.

(a) How many of the following statements about tendons are correct?

(1)

- tendons act as extensors
- tendons are myogenic
- tendons connect muscles to bones

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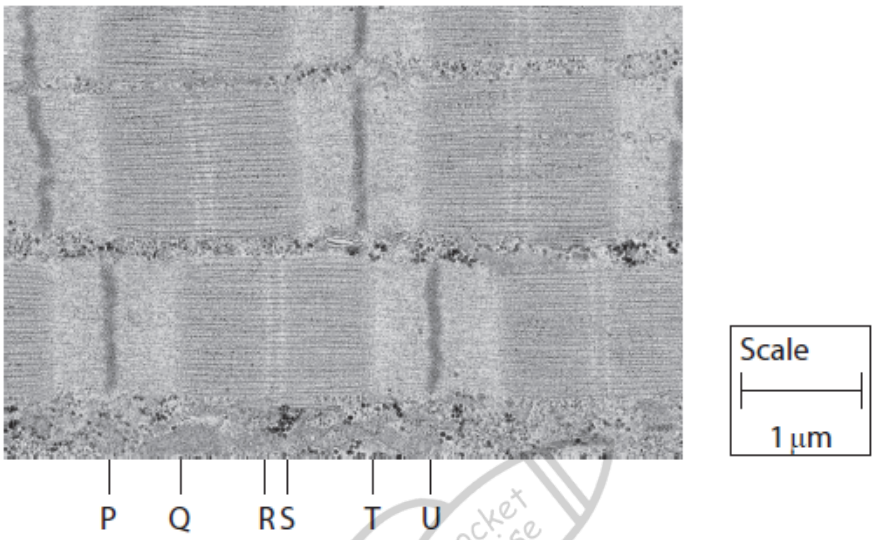
☒ **A** 0

☒ **B** 1

☒ **C** 2

☒ **D** 3

(b) The photograph shows part of a skeletal muscle fibre as seen using an electron microscope.



© THOMAS DEERINCK, NCMIR/SCIENCE PHOTO LIBRARY

(i) Which pair of letters represents the length of one sarcomere?

(1)

☒ **A** P to T

☒ **B** P to U

☒ **C** Q to T

☒ **D** R to S

(ii) Calculate the magnification of this image.

Give your answer in standard form.

(2)

Answer .....

(iii) Describe how the structure of a fast twitch muscle fibre differs from a slow twitch muscle fibre.

(4)

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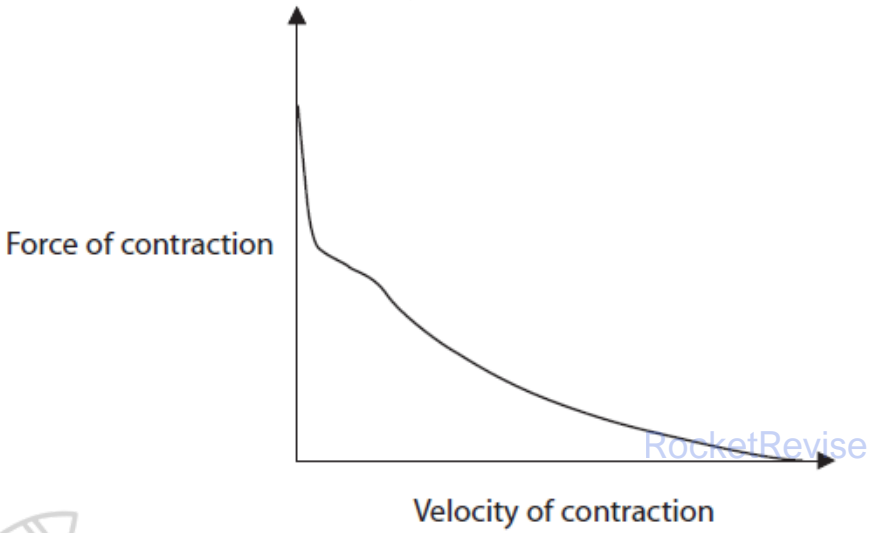
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(c) The graph shows the relationship between the force of muscle contraction and the velocity of muscle contraction.



The greater the number of cross bridges formed between actin and myosin the greater the force of contraction of a muscle.

Explain the relationship between the force and the velocity of contraction of a muscle.

(3)

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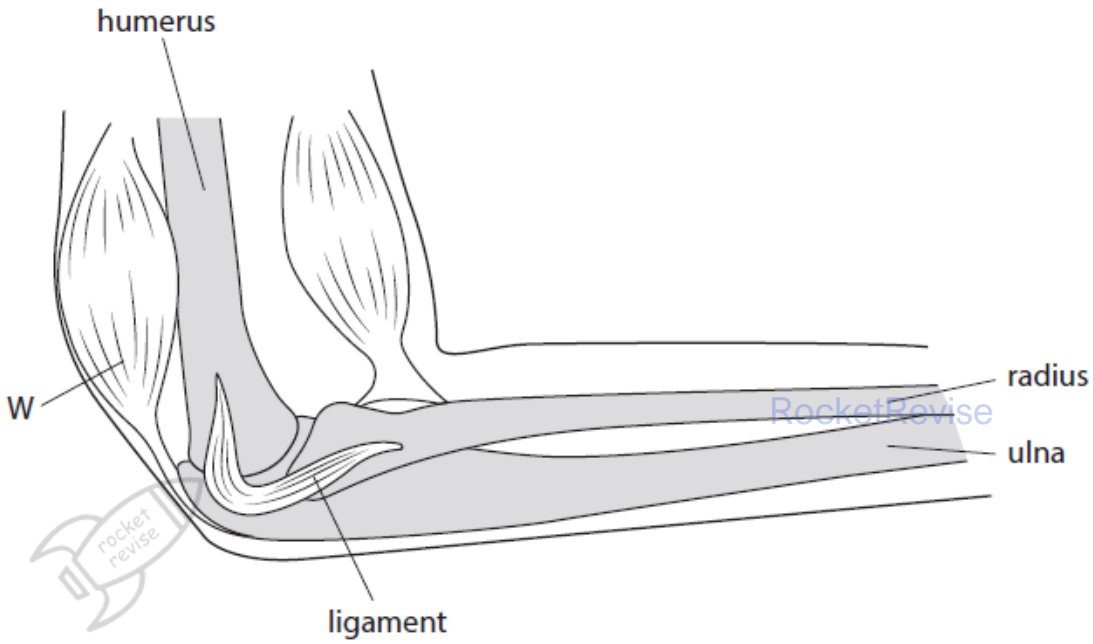
(Total for question = 11 marks)

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Q16.

Answer the question with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The diagram shows an elbow joint.



(a) (i) Which structure contains actin and myosin?

(1)

☒ **A** bone

☒ **B** ligament

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☒ **C** muscle

☒ **D** tendon

(ii) What is the function of the part of the joint labelled W?

(1)

☒ **A** it acts as an extensor

☒ **B** it acts as a flexor

☒ **C** it acts as a ligament

☒ **D** it acts as a tendon

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(b) (i) Explain the role of the ligament labelled on the diagram.

(2)

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(ii) Suggest **one** possible cause of damage to this ligament.

(1)

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(Total for question = 5 marks)

Q17.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new

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answer with a cross ☒.

Aerobic respiration is a metabolic process involving a series of chemical reactions.

(a) How is water formed during aerobic respiration?

(1)

☒ **A** reduction of carbon

☒ **B** reduction of oxygen

☒ **C** phosphorylation of ADP

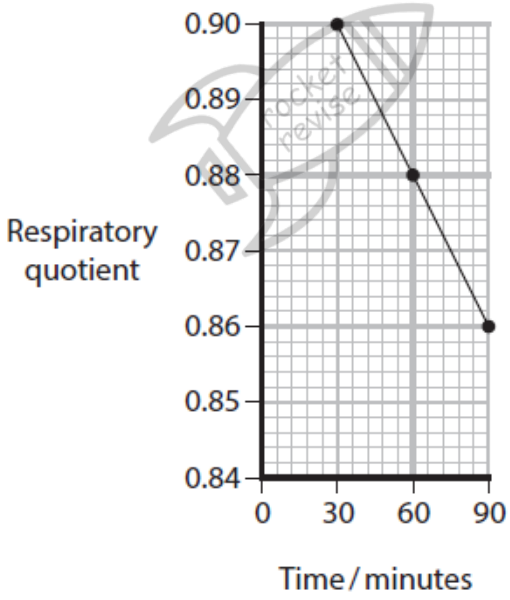
☒ **D** phosphorylation of glucose

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(b) The effect of exercise on the respiratory quotient of a cyclist was investigated.

The cyclist undertook moderate exercise for 90 minutes.

The graph shows the respiratory quotient for the cyclist during this investigation.



(i) The rate of oxygen uptake by the cyclist at 60 minutes was 2.20 dm^3 min^{-1}.

Which is the rate of carbon dioxide production by the cyclist at 60 minutes?

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(1)

☒ **A** 1.30 dm^3 min^{-1}

☒ **B** 1.94 dm^3 min^{-1}

☒ **C** 2.50 dm^3 min^{-1}

☒ **D** 3.08 dm^3 min^{-1}

(ii) The table gives the respiratory quotient for aerobic respiration using fats or carbohydrates as respiratory substrates.

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Substrate	Respiratory quotient
Fat	0.7
Carbohydrate	1.0

Explain the changes in respiratory quotient in this person during the exercise period of this investigation.

Use the information in the table and the graph to support your answer.

(2)

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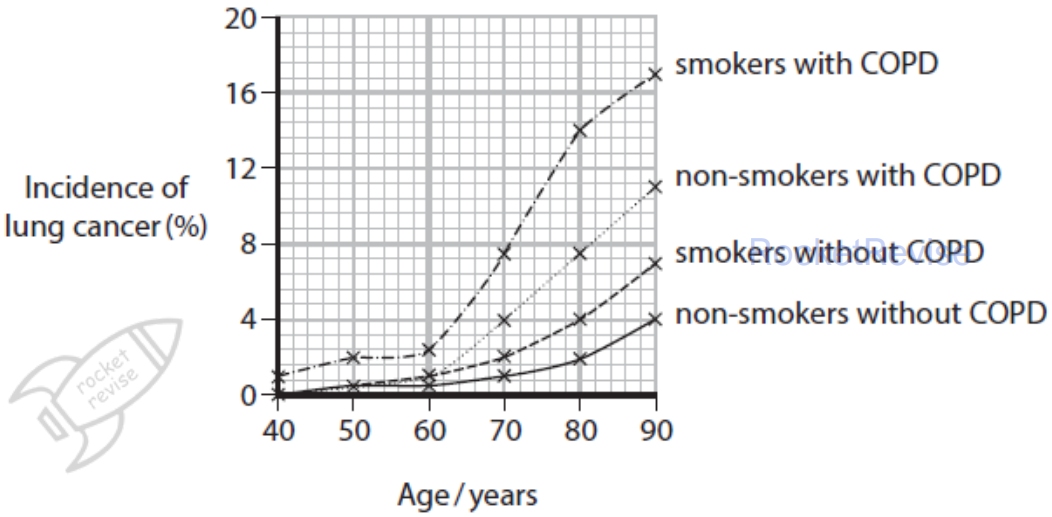
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Chronic obstructive pulmonary disease (COPD) is a life-threatening illness.

This disease greatly reduces the ability to take in oxygen in the lungs and is a risk factor for lung cancer.

The graph compares the incidence of lung cancer in smoking and non-smoking individuals with and without COPD.



In South Korea there are 105 500 males and 158 600 females in the 80-year-old age group.

(i) Calculate the total number of people in this age group that will have lung cancer who are non-smokers with COPD.

Give your answer in standard form.

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(2)

Answer .....

(ii) Cigarette smoking causes inflammation in the lungs. This can result in damage to the walls of the alveoli and lead to COPD.

Cigarette smoking also causes the production of thick mucus in the lungs.

Deduce the effects on the heart rate and ventilation rate in a person who smokes and has COPD.

(4)

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(Total for question = 10 marks)

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Q18.

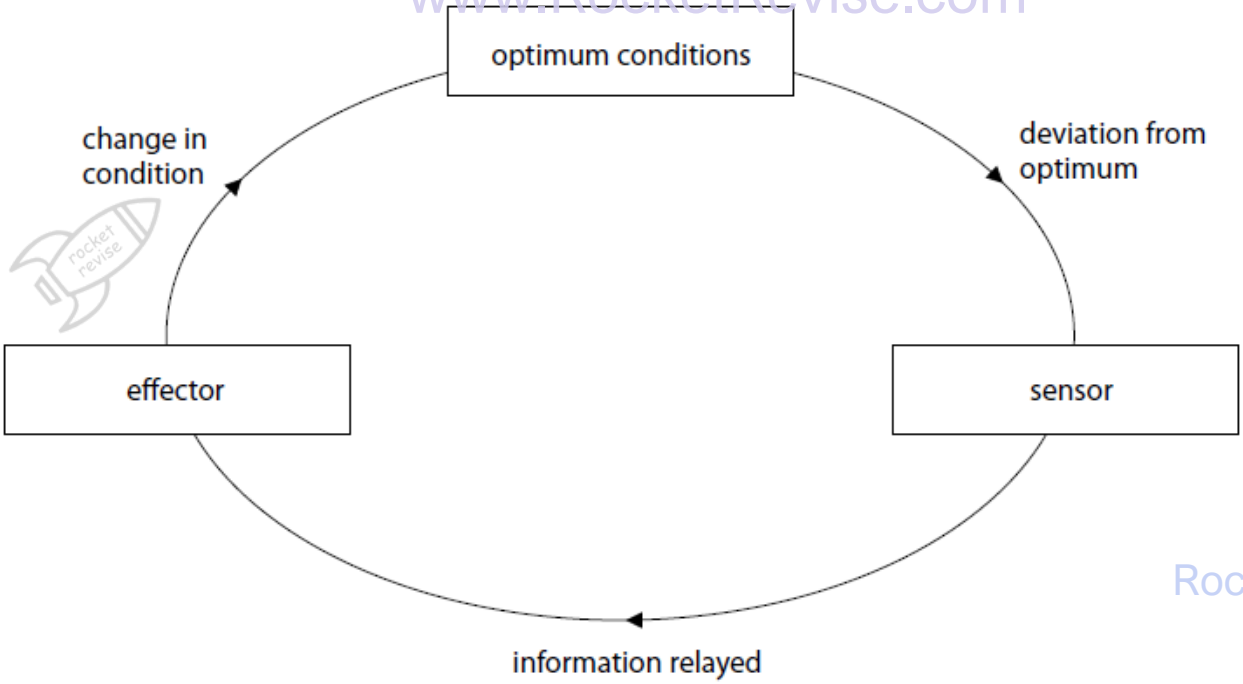
(a) Negative feedback is important in maintaining homeostasis.

Control of the ventilation rate is an example of negative feedback.

The diagram summarises negative feedback.

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(i) Which row of the table shows the sensor and the location of the sensor involved in the control of ventilation?

(1)

	sensor	location of sensor
☒ **A**	baroreceptor	hypothalamus
☒ **B**	baroreceptor	medulla oblongata
☒ **C**	chemoreceptor	hypothalamus
☒ **D**	chemoreceptor	medulla oblongata

(ii) Explain why the control of ventilation rate is an example of negative feedback.

Use the information in the diagram to support your answer.

(4)

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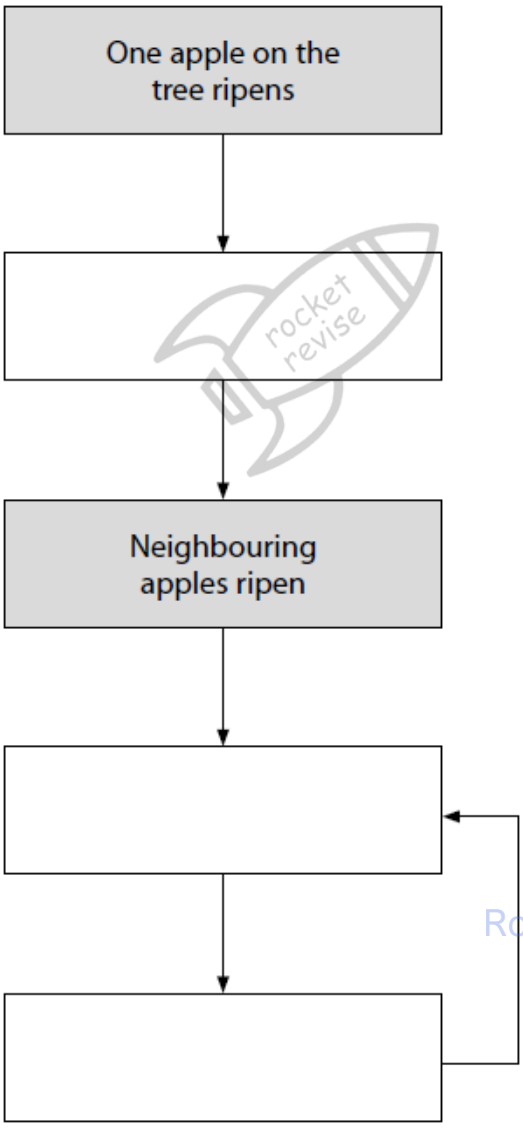
(b) The ripening of apples is an example of positive feedback.

Ethene is a gas that causes apples to ripen. Ripe apples release ethene.

Complete the diagram to show positive feedback in the ripening of apples.

(3)

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(Total for question = 8 marks)

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Q19.

Movement in mammals results from the interaction between muscles, tendons, ligaments and the skeleton.

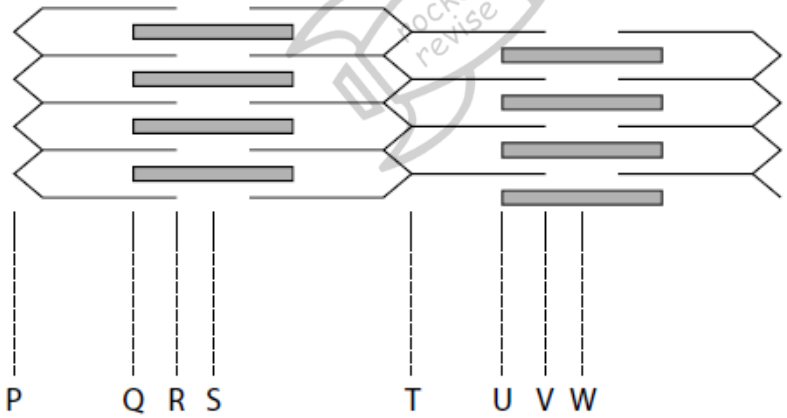
(a) Which row of the table describes tendons and ligaments?

(1)

	Tendons	Ligaments
☒ **A**	flexible, non-elastic tissue connecting bone to bone	flexible tissue connecting muscle to bone
☒ **B**	flexible, non-elastic tissue connecting muscle to bone	flexible tissue connecting bone to bone
☒ **C**	flexible tissue connecting bone to bone	flexible, non-elastic tissue connecting muscle to bone
☒ **D**	flexible tissue connecting muscle to bone	flexible, non-elastic tissue connecting bone to bone

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(b) (i) The diagram shows part of a skeletal muscle fibre.



Which pair of letters represents one sarcomere?

(1)

☒ **A** P to T

☒ **B** Q to U

☒ **C** R to V

☒ **D** S to W

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(ii) Compare and contrast the structure of fast-twitch and slow-twitch muscle fibres.

(4)

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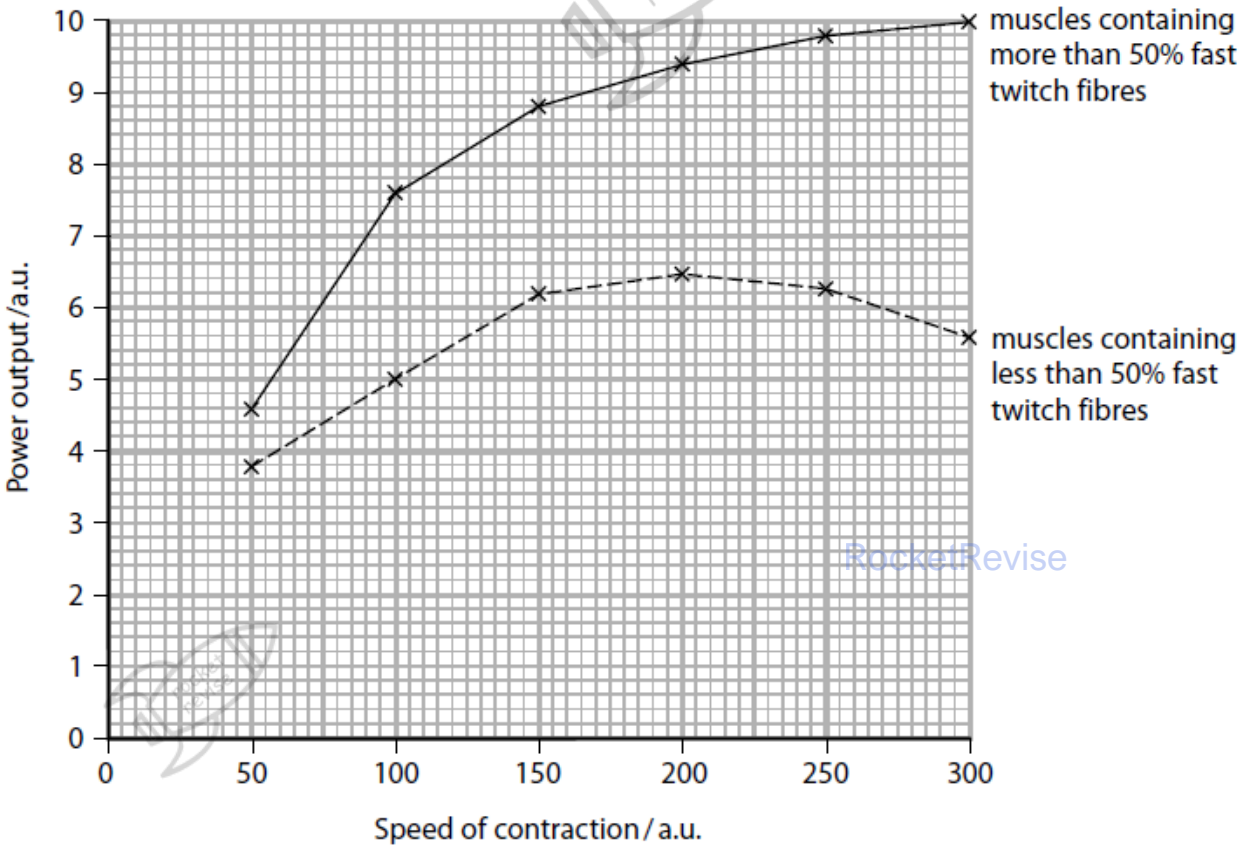
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*(c) The power output of a muscle depends on the speed of contraction and the proportion of fast-twitch and slow-twitch fibres in the muscle.

The graph shows the relationship between the power output and the speed of contraction of two different types of muscle.



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Explain the relationship between power output and the speed of contraction of these two types of muscle. Use the information in the graph and your own knowledge to support your answer.

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(Total for question = 12 marks)

Q20.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

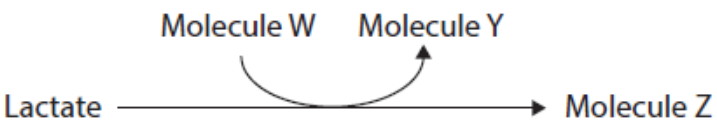
Mammals produce ATP by anaerobic and aerobic respiration.

(a) The effect of exercise intensity on the lactate concentration in human blood was investigated.

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- An athlete and a non-athlete performed the same exercise at the same intensity for three minutes.
- Their blood lactate concentrations were measured as soon as they finished exercising.
- This was repeated for different intensities of exercise.

(i) The diagram shows what happens to the lactate following exercise.



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Which row shows the molecules in this diagram?

(1)

	Molecule W	Molecule Y	Molecule Z
☒ **A**	oxidised NAD	reduced NAD	pyruvate
☒ **B**	oxidised NAD	pyruvate	reduced NAD
☒ **C**	reduced NAD	oxidised NAD	pyruvate
☒ **D**	reduced NAD	pyruvate	oxidised NAD

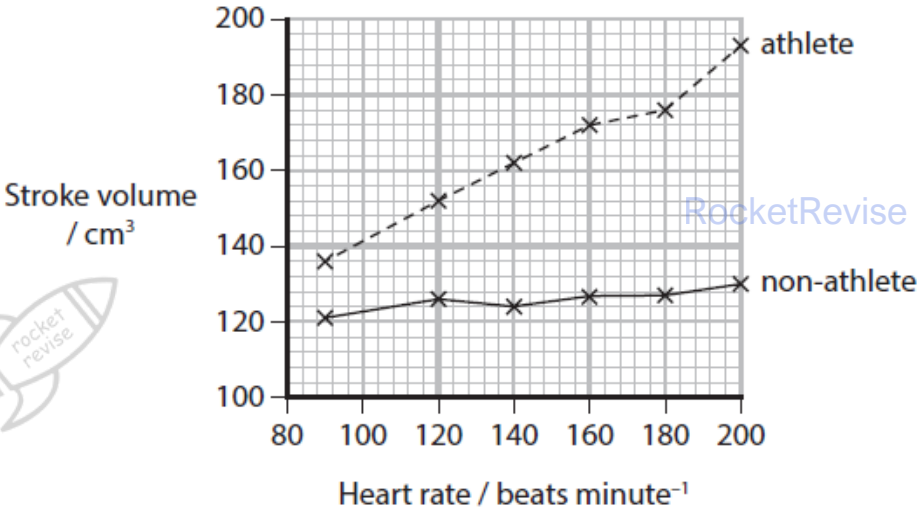
*(ii) The table and graph show some of the results of this investigation.

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Intensity of exercise / a.u.	Lactate concentration in blood / mmol dm^{-3}	
	Athlete	Non-athlete
0	1.0	1.5
60	1.1	2.2
120	1.4	3.7
180	3.2	6.2
240	6.4	10.0

Stroke volume is the volume of blood that leaves each ventricle during ventricular systole.

The graph shows the stroke volumes of the athlete and the non-athlete at different heart rates.



Explain the results of this investigation.
Use the information in the graph and the table and your own knowledge to support your answer.

(6)

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(b) The photograph shows a coyote.



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(Source: © Shaun Cunningham/Alamy Stock Photo)

Some coyotes have been poisoned with cyanide to prevent them from killing the livestock of farmers.

Cyanide inhibits cytochrome oxidase, the last carrier in the electron transport chain (ETC).

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Explain how the action of cyanide would kill a coyote.

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(Total for question = 11 marks)

Q21.

Endurance athletes may suffer from muscle fatigue when contraction of the muscle fibres is disrupted.

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It is thought that the disruption of the contraction of muscles is caused by a lack of calcium ions within the muscle fibre.

(a) Explain why a decrease in calcium ions could disrupt the contraction of muscles.

(3)

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(b) The body controls the concentration of calcium ions circulating in the blood.

This concentration is kept within the range 80 to 100 mg dm^{-3}.

(i) Name the type of process involved in maintaining calcium ion concentrations within this range.

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(1)

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(ii) The calcium ion concentration in the blood is kept within a range of 6% above and below the mean concentration.

The mean calcium ion concentration in the blood of a person was 91 mg dm^{-3}.

Calculate the **maximum** possible difference in the calcium ion concentration in the blood of this person.

(2)

Answer ..... mg dm^{-3}

(c) In an investigation, a group of cyclists completed a 30 km time trial in a simulator.

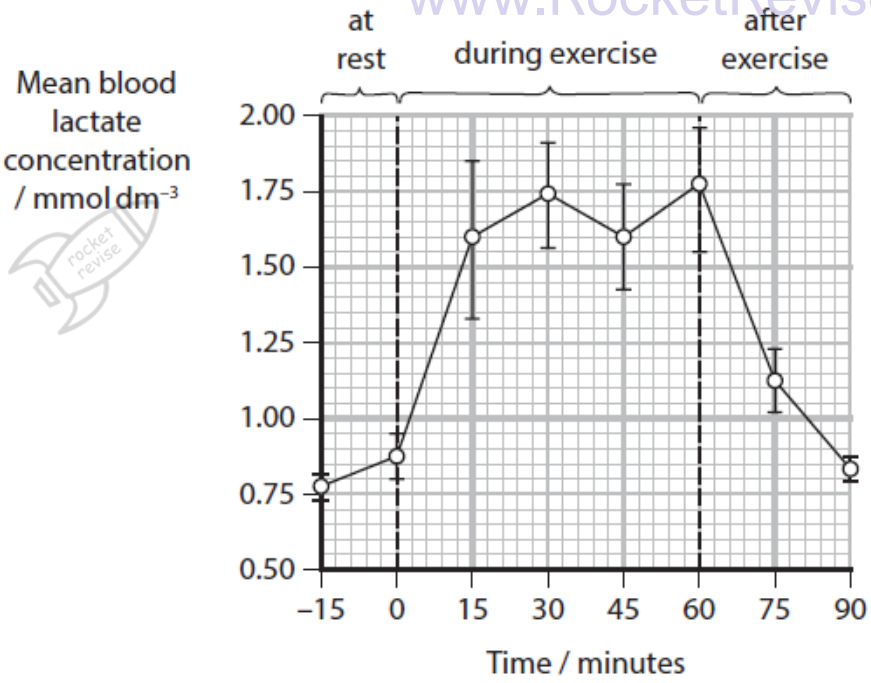
The blood lactate concentration was measured at rest, during exercise and after exercise.

The results are shown on the graph.

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(i) Explain the effect on the lactate concentration from 15 minutes to 60 minutes on the graph.

(2)

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(ii) State **one** reason why the lactate concentration was measured 15 minutes before starting the exercise.

(1)

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(iii) State **one** variable that would need to be controlled in this investigation and how this would be achieved.

(1)

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(Total for question = 10 marks)

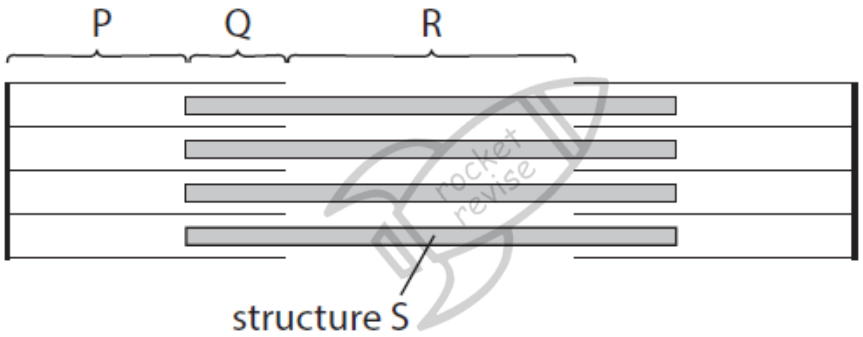
Q22.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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The sarcomere is the contractile unit of skeletal muscle fibres.

(a) The diagram shows a relaxed sarcomere.



(i) Which of the following would shorten when the sarcomere contracts?

(1)

☒ **A** region P only

☒ **B** region R only

☒ **C** regions P and R

☒ **D** regions Q and R

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(ii) A diagram of this sarcomere, drawn to the same scale, has a length of 50 mm when contracted.

Give the ratio of the lengths of the relaxed sarcomere to the contracted sarcomere.

(1)

Answer ..... : 1

(iii) Name the major protein component that forms structure S in the diagram.

(1)

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(b) Describe the role of troponin molecules in the contraction of a sarcomere.

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(c) Muscle contraction increases muscle tension.

The relationship between increasing ATP concentration and peak muscle tension was investigated.

The results are shown in the table.

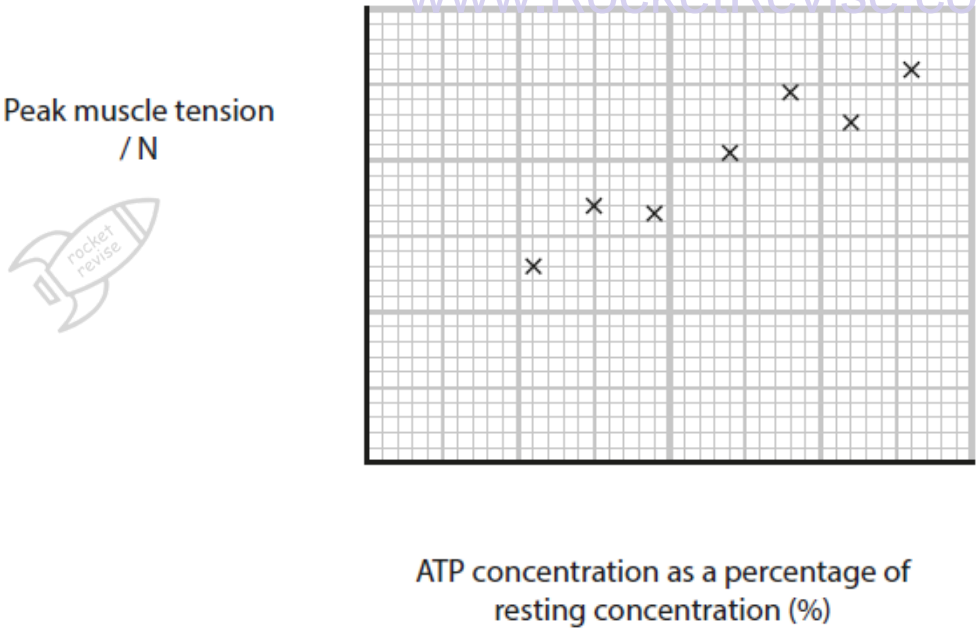
ATP concentration as a percentage of resting concentration (%)	Peak muscle tension / N
110	26
150	34
190	33
240	41
280	49
320	45
360	52

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The incomplete graph shows the results of this investigation.

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(i) Add an appropriate scale to the axes of the graph and draw a line of best fit.

(2)

(ii) Calculate the gradient of the line of best fit.

(1)

Answer ..... N per percentage increase

(Total for question = 10 marks)

Q23.

Exercise affects heart rate and blood pressure.

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(a) The effect of exercise on heart rate and blood pressure of a group of people was investigated.

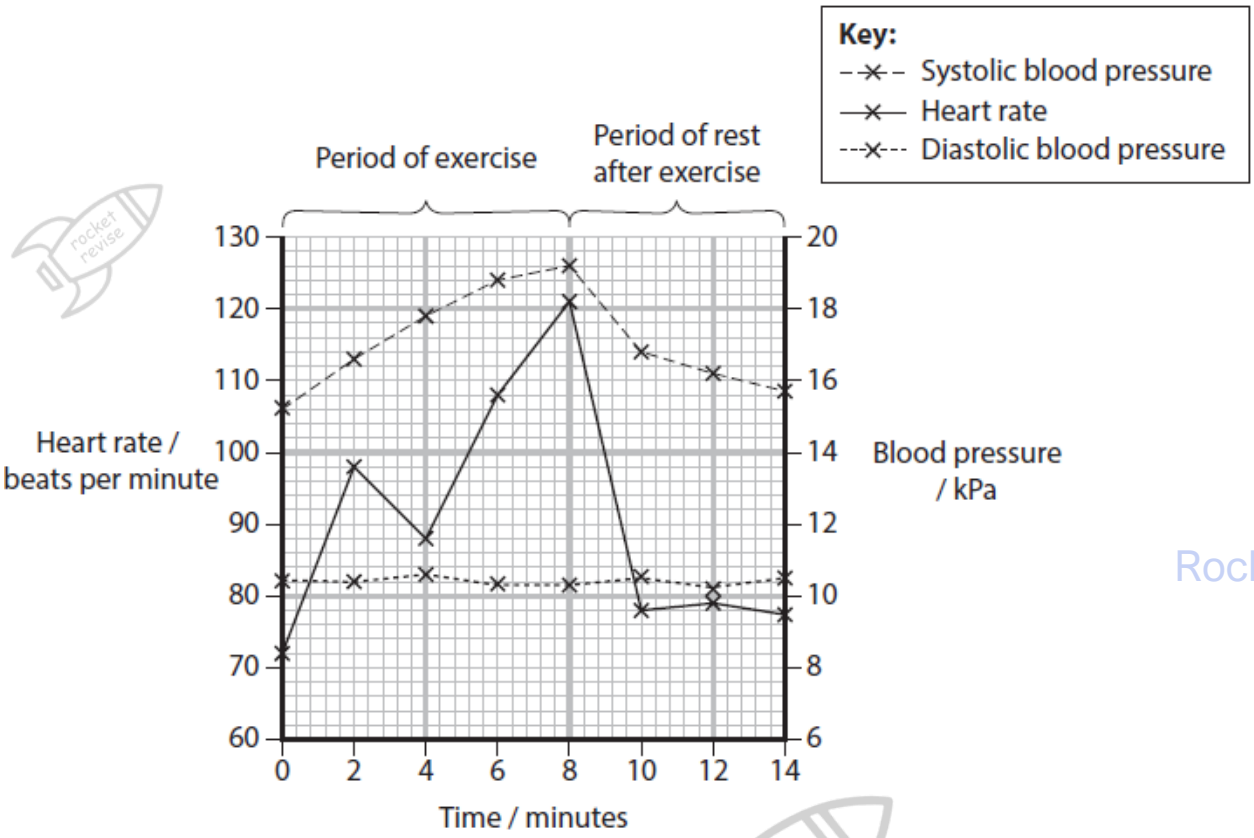
The mean heart rate and blood pressure were recorded:

- at the start of exercise
- every 2 minutes during the exercise
- every 2 minutes after the exercise.

The graph shows the results.

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(i) Describe **three** conclusions from this investigation.

(3)

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(ii) Describe how the systole and diastole parts of the cardiac cycle are controlled during the exercise.

(4)

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(b) At rest, a heart pumps 5 dm^3 $minute^{-1}$ of blood and has a heart rate of 72 beats $minute^{-1}$.

Calculate the stroke volume.

Use the equation:

cardiac output = heart rate × stroke volume

Give your answer in cm^3 and to **two** significant figures.

(2)

Answer ..... cm^3

(Total for question = 9 marks)

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